

# Le système international d'unités The International System of Units

9 édition 2019

Annexe 2

## Annex 2

Valeurs recommandées des fréquences étalons  
destinées à la mise en pratique de la définition du  
mètre et aux représentations secondaires de la  
seconde

**Recommended values of standard frequencies  
for applications including the practical  
realization of the metre and secondary  
representations of the second**

Partie 2

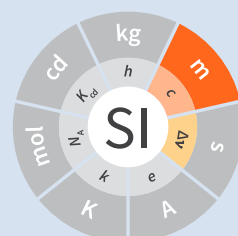
## Part 2

Ytterbium 171 Ion ()

**Ytterbium 171 Ion ()**

CCL-CCTF Frequency Standards Working Group

May 2016  
mai 2016





**Recommended values of standard  
frequencies for applications including  
the practical realization of the metre and  
secondary representations of the second  
Part 2: Ytterbium 171 Ion ( )**

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## 1. Recommended value [1] of the frequency

$$f(\text{Yb}^+, \text{quadrupole}) = 688\,358\,979\,309\,308.3 \text{ Hz}$$

equivalent to

$$\lambda(\text{Yb}^+) = 435\,517\,610.739\,688 \text{ fm}$$

with an estimated relative standard uncertainty of  $6 \times 10^{-16}$ .

This radiation is already endorsed by the CIPM as a secondary representation of the definition of the second.

## 2. Source data

Adopted value  $f(\text{Yb}^+) = 688\,358\,979\,309\,308.3\text{ Hz}$   
 $u_c/y = 6 \times 10^{-16}$

calculated from

| $f(^{171}\text{Yb}^+, \text{quadrupole}) / \text{Hz}$ | $u / \text{Hz}$ | $f(^{171}\text{Yb}^+ \text{ oct}) / f(^{171}\text{Yb}^+ \text{ quad})$ | $u$                   | source data |
|-------------------------------------------------------|-----------------|------------------------------------------------------------------------|-----------------------|-------------|
| 688 358 979 309 308                                   | 2.14            |                                                                        |                       | [2]         |
| 688 358 979 309 306.97                                | 0.73            |                                                                        |                       | [3]         |
| 688 358 979 309 310                                   | 9               |                                                                        |                       | [4]         |
| 688 358 979 309 308.42                                | 0.42            |                                                                        |                       | [5]         |
| 688 358 979 309 307.82                                | 0.36            |                                                                        |                       | [6]         |
|                                                       |                 | 0.932 829 404 530 964 65                                               | $3.1 \times 10^{-16}$ | [5]         |

as a weighted mean from the above values applying the method of Margolis and Gill [7]. Considering that the values came from only two laboratories, the CCTF considered it prudent to enlarge the uncertainty by a factor of two.

$^{171}\text{Yb}^+$  ion,  $6s\ ^2S_{1/2} - 5d\ ^2D_{3/2}$  unperturbed optical transition

## References

- [1] CIPM Recommendation 2 (CI-2015): Updates to the list of standard frequencies  
<http://www.bipm.org/jsp/en/CIPMRecommendations.jsp>
- [2] Report of the 12th meeting, Consultative Committee for Length (CCL), 15-16 Sept. 2005; available at <http://www.bipm.org/utis/common/pdf/CCL12.pdf> and Questionnaire previous to the 2006 meeting of the CCL/CCTF Joint Working Group CCL-CCTF/06-11.
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- [5] Godun, R. M.; Nisbet-Jones, P. B. R.; Jones, J. M.; King, S. A.; Johnson, L. A. M.; Margolis, H. S.; Szymaniec, K.; Lea, S. N.; Bongs, K. & Gill, P.: Frequency ratio of two optical clock transitions in  $^{171}\text{Yb}^+$  and constraints on the time-variation of fundamental constants, *Phys. Rev. Lett.*, **113**, 210801 (2014).
- [6] Tamm, C.; Huntemann, N.; Lipphardt, B.; Gerginov, V.; Nemitz, N.; Kazda, M.; Weyers, S. & Peik, E.: A Cs-Based Optical Frequency Measurement Using Cross-Linked Optical and Microwave Oscillators, *Phys. Rev. A*, **89**, 023820 (2014).
- [7] H. S. Margolis, P. Gill: Least-squares analysis of clock frequency comparison data to deduce optimized frequency and frequency ratio values; *Metrologia*, 2015, **52**, 628-634